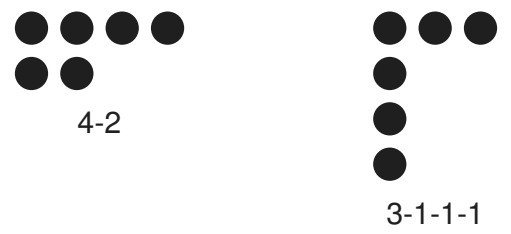


Problem 1: Lay 6 counters as rows, longest row on top, like the two pictures below. A move: take one counter from every pile, even from a pile of one, and put all the counters you took together as one new pile. Write a picture as its pile sizes, biggest first: 4-2, not 2-4. (a) Find every picture with 6 counters. Go in order of biggest pile, 6 first, and list them in the first column. (b) For each picture, make one move and write the picture one move later. (c) Draw the map on the next page: every picture inside a circle, and an arrow from each picture to the picture one move later. (d) Using the map, write for each picture how many moves it takes to reach the picture that gives itself back.



biggest pile	picture	one move later	moves to the picture that gives itself back
6			
5			
4			
3			
2			
1			

Number of pictures with 6 counters: _____

Problem 1, continued: Draw the map here. Put the picture that gives itself back near the bottom, so that the arrows flow down. Then answer (e), (f) and (g) under the box.



(e) Pictures that no arrow points into: _____

(f) The longest game with 6 counters starts at _____ and takes _____ moves. Write it as a chain of arrows:

(g) Could some start with 6 counters go on changing forever? Use the map to say how you know:

Problem 2: A picture is stuck if one move gives the same picture back. (a) Test each picture in the table: build it, make one move, write what you get, and write yes or no. (b) Find a stuck picture with 15 counters and one with 21 counters. Write them in the two empty rows and test them. (c) Try to build a stuck picture with 12 counters. Write what you tried and what went wrong. (d) A picture is stuck and its biggest pile has 5 counters. After the move, which pile of the new picture is the pile of 5, and how many piles must there be? Write the whole picture and check that it is stuck. (e) Say why 3-2-1 is stuck, pile by pile, and why the same reason works for a staircase with 10 rows. (f) List every stuck picture with at most 25 counters.

picture	one move later	stuck?
3-2-1		
2-2		
4-2		
3-3		
4-3-2-1		
6-3-1		
5-4-3-2-1		
4-4-2		

(c) With 12 counters I tried: _____

What went wrong: _____

(d) The pile of 5 after the move is _____, so there are _____ piles,
and the picture is _____.

(f) List every stuck picture with at most 25 counters, each with its number of counters:

Problem 3: Now use 8 counters. (a) Find every picture with 8 counters. Share the work: one of you takes biggest pile 5 or more, one takes biggest pile 4 or 3, one takes biggest pile 2 or 1. For each picture write the picture one move later. Count the pictures in all. (b) Follow the arrows from 8 until a picture repeats, and write that loop on the next page. (c) Follow the arrows from 2-2-2-2 until a picture repeats, and write that loop. (d) For every picture in the table, follow the arrows until you reach a loop, and mark the picture A or B for the loop you reach; if you reach a loop that is neither, write it down. (e) Draw the map on the next page: draw the two loops first, then attach the other pictures by their arrows. (f) Look for a rule that tells from the start alone which loop it will reach. Write your best rule and, if you find one, a start that breaks it.

biggest 5 to 8	one move later	A/B

biggest 2 or 1	one move later	A/B

biggest 4 or 3	one move later	A/B

Number of pictures with 8 counters: _____

Problem 3, the map. Loop A (from 8): _____

Loop B (from 2-2-2-2): _____



(f) My best rule for which starts reach loop B: _____

A start that breaks it: _____

Problem 4: Now use 10 counters. (a) Play from one pile of 10. (b) Play from ten piles of 1. (c) Play from 3-3-2-1-1. (d) Play from a start of your own. Write every picture and count the moves to the stuck picture. (e) Fill in the table: for 1, 3, 6 and 10 counters, the number of moves from a single pile, and the longest game you have found. (f) Guess both numbers for 15 counters and write them in the table.

A: 10		B: ten piles of 1		C: 3-3-2-1-1		D: own start	
Start		Start		Start		Start	
1		1		1		1	
2		2		2		2	
3		3		3		3	
4		4		4		4	
5		5		5		5	
6		6		6		6	
7		7		7		7	
8		8		8		8	
9		9		9		9	
10		10		10		10	
11		11		11		11	
12		12		12		12	
13		13		13		13	

Moves: A _____ B _____ C _____ D _____

counters	moves from one pile	longest game found	its start
1			
3			
6			
10			
15			

Problem 4, continued: (f, continued) Play from one pile of 15 in the column headed “15, one pile”, numbers only, and check your guess in the table. (g) Look for a start with 10 counters that takes 11 moves or more. Use the other three columns, and write each start you try in its Start box.

15, one pile		E: 10 counters		F: 10 counters		G: 10 counters	
Start		Start		Start		Start	
1		1		1		1	
2		2		2		2	
3		3		3		3	
4		4		4		4	
5		5		5		5	
6		6		6		6	
7		7		7		7	
8		8		8		8	
9		9		9		9	
10		10		10		10	
11		11		11		11	
		12		12		12	
		13		13		13	

Moves from one pile of 15: _____ Moves: E _____ F _____ G _____

The longest game with 10 counters I found starts at _____ and takes _____ moves.

Problem 5: (a) Play the six games below until a picture repeats, numbers only; “8 from 2-2-2-2” means 8 counters starting at 2-2-2-2. Play from one pile of 4 and of 5 the same way, straight into the table on the next page. Write every loop, in order, and its length in that table.

7 from 7	
Start	
1	
2	
3	
4	
5	
6	
7	
8	

8 from 8	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	

8 from 2-2-2-2	
Start	
1	
2	
3	
4	
5	
6	

9 from 9	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

11 from 11	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	

12 from 12	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Problem 5, continued: Fill in the table from your games. (b) Draw the four loop pictures for 7 counters as rows of dots in the boxes, in order. In each drawing, find a 3-2-1 stairs, color it, and put a ring around every dot left over. (c) Do the same for the five loop pictures with 11 counters, looking for a 4-3-2-1 stairs inside each one. (d) Do the same for the four loop pictures with 9 counters, with a 3-2-1 stairs. Tell the adult how the ringed dots move from one picture to the next.

counters	start	loop pictures, in order	length
4	4		
5	5		
7	7		
8	8		
8	2-2-2-2		
9	9		
11	11		
12	12		

(b)

1	2	3	4
---	---	---	---

(c)

1	2	3	4	5
---	---	---	---	---

(d)

1	2	3	4
---	---	---	---

Problem 6: Pretend 4-2-1 was just made by a move and undo it: one of its piles is the new pile. That pile gave one counter to every other pile, and any counters it had left over were piles of 1 that vanished in the move. (a) Try each pile of 4-2-1 as the new pile. Find every picture that leads to 4-2-1 in one move. (b) Find every picture that leads to 5-3-2. (c) Find every picture that leads to 3-2-1. (d) Find every picture with 8 counters that no picture leads to, and check against your table or map from Problem 3. (e) Decide without playing whether some picture leads to each picture in the small table. (f) Tell the adult how you decide.

(a) Pictures that lead to 4-2-1: _____

(b) Pictures that lead to 5-3-2: _____

(c) Pictures that lead to 3-2-1: _____

(d) Pictures with 8 counters that nothing leads to: _____

picture	something leads to it?	which pile was the new pile
3-3-3		
2-2-2-2-1		
5-1-1-1-1-1-1		
4-4-4-4-4		
3-1-1-1-1-1-1		
9-1-1-1-1-1-1-1-1-1		

Problem 7: How many pictures are there with 1, 2, 3, 4, 5, 6, 7 and 8 counters? You already know two of these numbers. Write your lists for 4, 5 and 7 counters in the box, each list in order of biggest pile, and fill in the table. Then tell the adult how you made sure your list for 7 counters is complete.

counters	1	2	3	4	5	6	7	8
pictures								

Problem 8: Use playing cards as counters. Lay them as in the first grid: row 1 is A, 2, 3, 4; row 2 is 2, 3, 4, 5; row 3 is 3, 4. (a) Check every card: its rank (A counts as 1) equals its row number plus its column number minus 1. (b) Make a move in this special way: take the left column, and lay it as a new top row, with the card from row 1 in column 1, the card from row 2 in column 2, and so on; slide every other card one column to the left; do not reorder the rows. Write the cards after each move in grids 1 to 4. After each move, check (a) again. (c) Make a fifth move the same way and write it in grid 5. Now the rows are out of order. Sort them longest on top by swapping neighbouring rows, and write the result in grid 6. Which cards now break the rule in (a)? (d) For grid 5 and for grid 6, add up row plus column minus 1 over all ten cards. Which total is smaller, and by how much?

Start: 4-4-2

A	2	3	4	
2	3	4	5	
3	4			

1

2

3

4

5

6 (sorted)

(c) Cards that break the rule after sorting: _____

(d) Grid 5 total: _____ grid 6 total: _____ difference: _____

Problem 8, continued: (e) Start again with 7-1-1-1, laid as in the start grid: row 1 is A, 2, 3, 4, 5, 6, 7; row 2 is 2; row 3 is 3; row 4 is 4. Check that every card obeys the rule in (a). Make one move the special way and write it in grid 7. Sort the rows longest on top and write grid 8. Which cards break the rule now? Add up row plus column minus 1 over all ten cards for grid 7 and for grid 8.

Start: 7-1-1-1

A	2	3	4	5	6	7
2						
3						
4						

7

8 (sorted)

(e) Cards that break the rule after sorting: _____

Grid 7 total: _____ grid 8 total: _____ difference: _____

The two rows swapped to make grid 6 had _____ and _____ cards; total dropped by _____. The two rows swapped to make grid 8 had _____ and _____ cards; total dropped by _____.

Problem 9: Now use 12 counters, numbers only. (a) Play from one pile of 12 and from 2-2-2-2-2 until a picture repeats. Write both loops. (b) Draw the pictures of both loops as rows of dots. In each drawing, find a 4-3-2-1 stairs, color it, and ring the dots left over. (c) Look for a start with 12 counters whose loop has a length other than 5. Play these on the large sheet, numbers only. Write the starts you tried and the loop length each one reached in the small table.

A: 12	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	

B: 2-2-2-2-2	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	

(c) start	loop length

Loop A: _____

Loop B: _____

(b)

A1	A2	A3	A4	A5
B1	B2	B3	B4	B5